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Conference Paper · November 2019

DOI: 10.1109/HNICEM48295.2019.9072892

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Tomayto, Tomahto: A Machine Learning Approach for Tomato Ripening Stage Identification Using Pixel-Based Color Image Classification

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Abstract—The main enterprise of the Philippine agriculture sector is crop cultivation where tomato is deliberated as one of the major crops in the country. With the abundance on tomato production, ripeness classification becomes fairly laborious and challenging, not to mention the subjective visual interpretation of human graders grounded from practical experience that is easily influenced by the environment and prone to error. Thus, this study proposes an automatic tomato ripeness identification using Support Vector Machine (SVM) classifier and CIELab color space via a machine learning approach. Dataset used for modeling and validation experiment in a 5-fold cross-validation strategy was composed of 900 images assembled from a farm and various image search engines. Divided into six classes that represent tomato ripening stages, experimental results showed that the proposed method was successful with 83.39% accuracy in ripeness classification detection. With this machine learning approach and combination of image processing techniques, the agriculture industry could benefit by automating the ripeness estimation which then could save tomatoes from damage.

Keywords— Image Processing, Image Classification, Support Vector Machine, Machine Learning, CIELAB color space

I. INTRODUCTION

Despite the transformation to an industrialized economy, Philippines is still primarily an agricultural country [1] where the gross value of agricultural production is amounted to PhP 429.7 billion during the first quarter of 2019 [2]. Employing 39.8% of the country's labor force [3], the main enterprise of the agriculture sector of the Philippines is crop cultivation [4] such as tomatoes, rice, corn, coconut, sugarcane, pineapple, mango, banana, and abaca, just to name a few. According to Philippine Statistics Authority, tomatoes are one of the major crops in the country [5]. In fact, tomato production reached 95.30 thousand metric tons from January to March 2019 [6]. The tomato (*Solanum lycopersicum*), known as “Kamatis” in the Philippines, has attracted the interest of food markets due to its medicinal impact such as decreasing the risk of various health conditions such as cancer, cardiovascular disease, and osteoporosis, and nutritive values such as phosphorus, iron, calcium, and vitamin C [7]. Lycopene, the major carotenoid in the fruit, is only amassed during the final ripening stage which accounts for 80% of the total carotenoid content [8]. Hence, tomato ripeness estimation has been viewed as a vital process that influences its quality evaluation. In fact, consumers use ripeness as a strand of what defines quality fresh fruits, which is grounded from its visual appearance such as color, size, and shape. Fruits color skin is not only an important determinant

of fruit selling price, but also as a feature of fruit ripeness [9] for numerous agricultural products such as apples [10, 11], bananas [12, 13], lime [14], mangoes [15-17], tomatoes [18, 19], and watermelons [20]. The abundance of harvest made it challenging to consistently determine tomato ripeness, which then becomes a problem when exported to a far place. Besides, farmers and human graders are usually subjective in ripeness visual interpretation grounding it from practical experience and/or ripeness classification charts, and easily influenced by the environment and prone to mistakes [21]. Thus, tomatoes are either unripe or overripe when they arrive at the market. This is the primary reason why tomatoes are often harvested during the “green” stage (See Fig. 1) to endure transportation [22]. Nevertheless, consumers are still less likely to purchase tomatoes when they are not on the “red” stage of ripening.

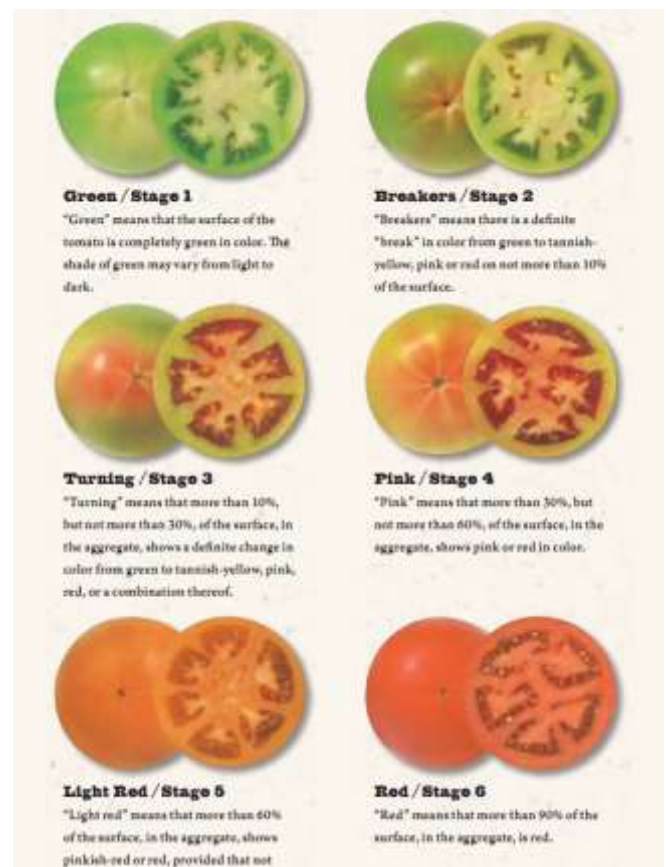


Fig. 1. Maturity and Ripening Stages of Tomatoes Based on United States Standards for Grades of Fresh Tomatoes.

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2. Materials and Methods

2.1. Experimental Design and Data Collection

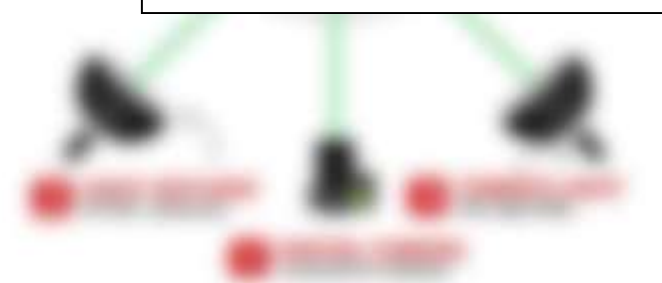
The study was conducted in a controlled laboratory environment. A total of 20 participants, consisting of 10 males and 10 females, were recruited from a local university. All participants were right-handed and had no history of neurological or psychiatric disorders. They were all naive to the purpose of the study. The experiment was approved by the local ethics committee. Participants were seated in a dimly lit room and viewed the stimuli through a computer monitor. The distance between the participant and the monitor was 57 cm. The stimuli were presented at the center of the screen. The background was black. The stimuli were presented for 100 ms. The inter-trial interval was 200 ms. The experiment was controlled using MATLAB (The MathWorks Inc., Natick, MA, USA) and the Psychtoolbox (Brain Research Laboratories, University of California, San Diego, CA, USA).

2.2. Stimulus Presentation and Task

The stimuli were presented on a computer monitor. The stimuli were presented for 100 ms. The inter-trial interval was 200 ms. The experiment was controlled using MATLAB (The MathWorks Inc., Natick, MA, USA) and the Psychtoolbox (Brain Research Laboratories, University of California, San Diego, CA, USA). The stimuli were presented at the center of the screen. The background was black. The stimuli were presented for 100 ms. The inter-trial interval was 200 ms. The experiment was controlled using MATLAB (The MathWorks Inc., Natick, MA, USA) and the Psychtoolbox (Brain Research Laboratories, University of California, San Diego, CA, USA).

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2.3. Data Analysis and Statistical Methods

The data were analyzed using MATLAB (The MathWorks Inc., Natick, MA, USA) and the Psychtoolbox (Brain Research Laboratories, University of California, San Diego, CA, USA). The data were analyzed using MATLAB (The MathWorks Inc., Natick, MA, USA) and the Psychtoolbox (Brain Research Laboratories, University of California, San Diego, CA, USA). The data were analyzed using MATLAB (The MathWorks Inc., Natick, MA, USA) and the Psychtoolbox (Brain Research Laboratories, University of California, San Diego, CA, USA).

2.4. Results and Discussion

The results of the study are presented in this section. The data were analyzed using MATLAB (The MathWorks Inc., Natick, MA, USA) and the Psychtoolbox (Brain Research Laboratories, University of California, San Diego, CA, USA). The data were analyzed using MATLAB (The MathWorks Inc., Natick, MA, USA) and the Psychtoolbox (Brain Research Laboratories, University of California, San Diego, CA, USA). The data were analyzed using MATLAB (The MathWorks Inc., Natick, MA, USA) and the Psychtoolbox (Brain Research Laboratories, University of California, San Diego, CA, USA).

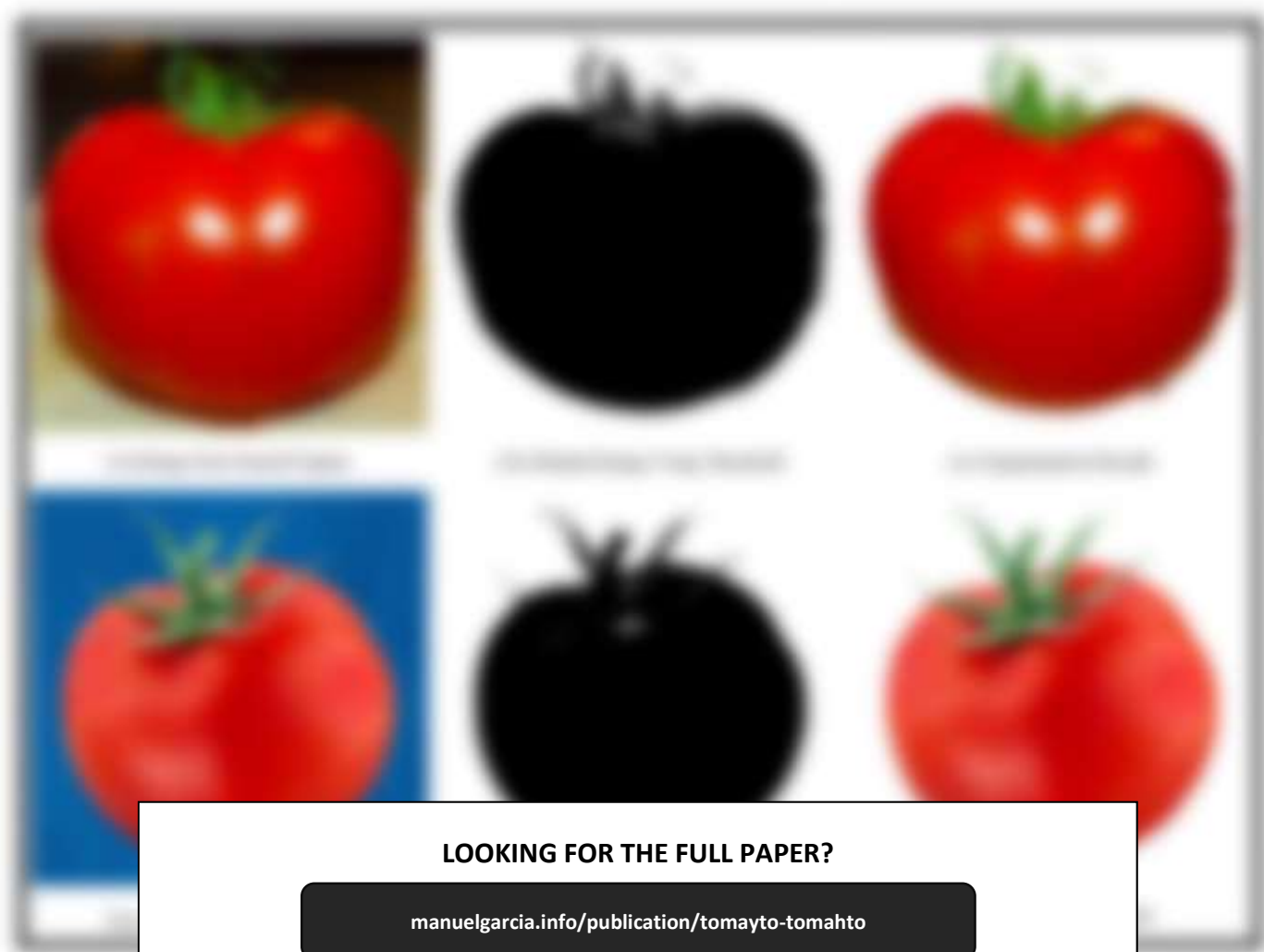


Figure 1: Tomatoes and their corresponding color maps.

The figure displays six images of tomatoes arranged in a 2x3 grid. The top row shows a red tomato on a light background, a black tomato silhouette on a white background, and a red tomato on a white background. The bottom row shows a red tomato on a blue background, a black tomato silhouette on a white background, and a red tomato on a white background. A white box with black text is overlaid on the bottom row, providing information about the full paper and a link to the publication.

Figure 2: Heatmap visualization of the tomato color map.

The figure shows a heatmap visualization of the tomato color map. The heatmap is a square image with a color scale on the right side. The color scale ranges from 0 (dark red) to 1 (green). The heatmap shows a diamond-shaped region of green color in the center, surrounded by a red region. The color scale is labeled with values from 0 to 1.

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